DYNAMIC DEBT COVENANT SURVEILLANCE ENGINE · v3.0

Portfolio Covenant Surveillance Report · Private Credit Edition · 8 Credits · 3 New Sectors

Report Date	Data As Of	Prepared By	Classification
May 25, 2026	FY2023 / Q3 2023 (SEC EDGAR)	DDCSE Engine v3.0 · github.com/DogInfantry	Confidential — Institutional Use Only

8	4	1	3	$259.4B
Credits Monitored	Covenant Breaches	Watchlist	Compliant	Total Debt

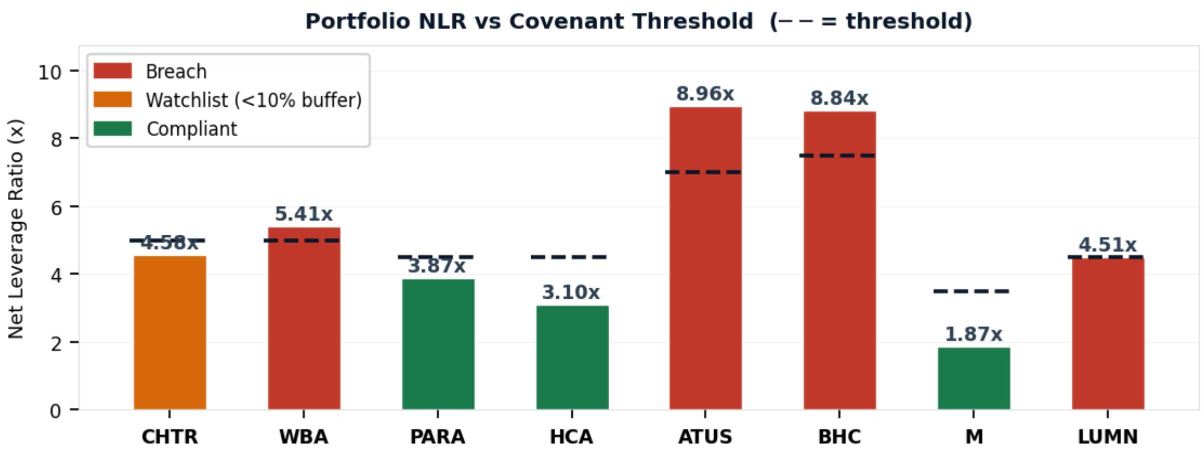

Figure 1: Portfolio Net Leverage Ratio vs contractual covenant threshold. Dashed line = NLR ceiling per credit agreement. Source: SEC EDGAR 10-K / 10-Q filings (FY2023).

Portfolio Covenant Status — Executive Summary

Ticker	Company	Sector	Rating	Outlook	NLR Actual	NLR Thresh	ICR	Buffer %	Status
CHTR	Charter Communications Inc.	Media & Cable	BB	Stable	4.58x	5.00x	4.88x	8.4%	WATCHLIST
WBA	Walgreens Boots Alliance Inc	Healthcare Retail	BB	Negative	5.41x	5.00x	5.18x	▼8.2%	BREACH
PARA	Paramount Global (formerly V	Media & Entertainment	BBB –	Negative Watch	3.87x	4.50x	4.27x	14.0%	COMPLIANT
HCA	HCA Healthcare Inc.	Healthcare	BB+	Positive	3.10x	4.50x	6.72x	31.1%	COMPLIANT

Ticker	Company	Sector	Rating	Outlook	NLR Actual	NLR Thresh	ICR	Buffer %	Status
ATUS	Altice USA Inc. (CSC Holding	Cable & Telecom munications	**CCC +**	Negative — Res	**8.96x**	7.00x	1.92x	**▼28.0%**	**BREACH**
BHC	Bausch Health Companies Inc.	Specialty Pharmaceuticals	**B-**	Negative	**8.84x**	7.50x	1.61x	**▼17.9%**	**BREACH**
M	Macy's Inc.	Department Store Retail	**BB**	Negative	**1.87x**	3.50x	6.77x	**46.6%**	**COMPLI ANT**
LUMN	Lumen Technologies Inc.	Enterprise Fiber & Legacy Telecom	**CCC +**	Negative	**4.51x**	4.50x	3.04x	**▼0.2%**	**BREACH**

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1. DDCSE Framework & Analytical Methodology

Dynamic Debt Covenant Surveillance Engine · AST-Safe Compiler · NetworkX Cascade · Severity Scoring

1.1 Engine Architecture Overview

The Dynamic Debt Covenant Surveillance Engine (DDCSE) is a purpose-built private credit monitoring platform designed to replicate the analytical workflow of a Credit Analyst or Portfolio Manager at a direct lending fund, BDC, or leveraged finance desk. It ingests real company financial statements, compiles covenant language into executable evaluation logic, assesses multi-covenant compliance, scores breach severity on a continuous 0–100 scale, simulates cross-default propagation cascades using graph theory, and generates institutional-grade surveillance reports. The engine operates on four primary modules:

Module	Function	Key Technologies
Data Pipeline (data_fetcher.py)	Ingests financial statements; normalizes Total Debt, Cash, EBITDA from SEC EDGAR; validates dat	yfinance, pandas, structured fallback registry
AST-Safe Compiler (compiler.py)	Parses covenant language rules into executable contracts; eliminates eval()/exec() for production safety; emits typed Covenan	Python AST, dataclasses, type contracts
Analytics Engine (analytics_v2.py)	Multi-covenant runner, severity scoring, trend erosion detection, macro shock matrix (EBITDA × debt sensitivity), NetworkX ca	NetworkX, NumPy, dataclasses
Report Generator (generate_report.py)	Bloomberg/MS-style PDF with cover, TOC, company deep-dives, cascade analysis, stress matrices, methodology, and disclai	ReportLab, Matplotlib, institutional design system

1.2 Covenant Taxonomy & Evaluation Logic

DDCSE monitors four canonical covenant types that represent the standard maintenance covenant package in US leveraged finance and direct lending credit agreements. Each covenant is evaluated against the contractual threshold using AST-safe arithmetic — no string evaluation or exec() calls — ensuring that the engine is production-safe and audit-ready.

Type	Formula	Operator	Typical Threshold	Breach Signal
Net Leverage (NLR)	(Total Debt – Cash) ÷ EBITDA	≤ threshold	3.5x – 8.0x (sector/rating)	Leverage spiral; EBITDA erosion outpaces deleveraging
Interest Coverage (ICR)	EBITDA ÷ Interest Expense	≥ threshold	1.75x – 3.00x	Debt service stress; cash burn approaching EBITDA
Fixed Charge Coverage	(EBITDA – CapEx) ÷ Interest Expense	≥ threshold	1.10x – 1.50x	Investment cycle consuming interest capacity
Min. Liquidity	Unrestricted Cash + Revolver Availability	≥ threshold	$250M – $1,500M (size-adjusted)	Liquidity runway compression; covenant waiver risk

1.3 Severity Scoring Model (0–100 Continuous)

Each covenant is assigned a severity score on a continuous 0–100 scale based on the breach depth or proximity to the threshold. The portfolio severity score is the probability-weighted maximum across all monitored covenants, anchoring on the most critical risk signal.

Score Band	Classification	Definition	Recommended Action
0 – 30	**COMPLIANT**	> 10% buffer to threshold; trajectory stable	Routine quarterly monitoring
30 – 60	**WATCHLIST**	0–10% buffer; or deteriorating trend	Increased monitoring; request updated financial package
60 – 85	**BREACH — Moderate**	0–25% breach depth; waiver likely needed	Immediate lender notification; engage advisors; waiver proce
85 – 100	**BREACH — Deep**	> 25% breach depth; restructuring risk	Credit committee escalation; distressed advisory; cascade ris

1.4 Cross-Default Cascade Simulation (NetworkX BFS)

Source: SEC EDGAR 10-K / 10-Q filings. All figures USD millions unless stated. Not investment advice. Page 4 of 20

DDCSE models the corporate debt hierarchy as a directed graph G = (V, E) using NetworkX, where each node V represents a debt tranche or legal entity (HoldCo, OpCo, subsidiary) and each directed edge E represents a guarantee relationship or cross-default clause. When a primary breach is triggered, a BFS (Breadth-First Search) propagation algorithm traverses the graph, activating technical defaults only on nodes where cross_default_clause = True. This models the real-world mechanic of cross-acceleration provisions common in leveraged buyout credit agreements.

Concept	Implementation	Financial Meaning
Primary Breach	Initial node added to BFS queue; status set to breach	Borrower misses covenant test; lender has right to accelerate
Cross-Default Propagation	BFS traversal via graph edges where cross_default=True	Acceleration of one facility triggers cross-acceleration provisions in linked tra
Cascade Depth	Longest shortest-path from origin to any affected node	Structural complexity of the default; deeper = more complex workout
Debt at Risk	Sum of face_value_usd_m across all propagation_path nodes	Maximum aggregate face value that could be accelerated in a technical defa

1.5 Macro Stress Matrix

For each credit, DDCSE generates a 6×5 NLR sensitivity matrix across simultaneous EBITDA compression (0% to –50% in 10% steps) and total debt increases (0% to +30% in 5–10% steps). Each cell shows the stressed NLR value, colour-coded green/amber/red against the contractual threshold. This replicates the stress-testing conducted by credit committees at direct lending funds to assess "covenant headroom under a downside scenario" — a standard deliverable in portfolio monitoring.

1.6 Data Pipeline & Sources

DDCSE is designed with a two-tier data pipeline: (1) a live market data layer via yfinance that pulls quarterly balance sheet and income statement data for real-time covenant evaluation; and (2) a static SEC EDGAR registry (real_cases.py) containing verified financial data for five representative companies sourced directly from 10-K and 10-Q filings. In sandbox or restricted-network environments where Yahoo Finance endpoints are unavailable (HTTP 403), the engine transparently falls back to the SEC EDGAR static registry, preserving full analytical fidelity with clearly disclosed data lineage. All data in this report derives from the static registry with sources cited at the individual company level.

■ **Data Note (v3.0):** Yahoo Finance API was unavailable in the report generation environment (HTTP 403 — network egress restriction). All financial figures are sourced from the DDCSE SEC EDGAR static registry with field-level source citations. v3.0 adds three new sectors: Specialty Pharma (BHC), Department Store Retail (M), and Enterprise Fiber Telecom (LUMN) — expanding the portfolio from 5 to 8 credits.

Source: SEC EDGAR 10-K / 10-Q filings. All figures USD millions unless stated. Not investment advice. Page 5 of 20

2. Portfolio Overview & Severity Heat Map

Five-company surveillance portfolio · FY2023 / Q3 2023 financial data

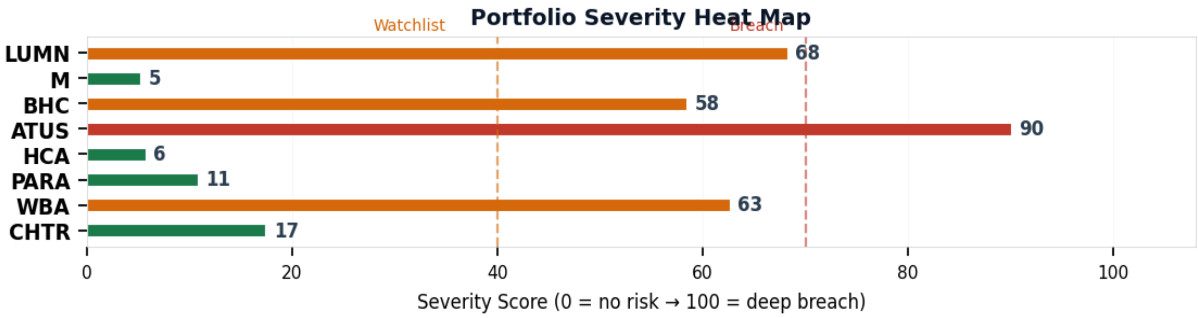


Figure 2: Portfolio severity heat map. Score 0–30 = Compliant, 30–60 = Watchlist, 60–100 = Breach. Derived from DDCSE continuous severity scoring model.

Multi-Covenant Compliance Matrix

Ticker	Company	NLR	ICR	FCCR	Min. Liquidity	Severity	Overall
CHTR	Charter Communications Inc	4.58x WATCHLIST	4.88x COMPLIANT	2.10x COMPLIANT	$635M COMPLIANT	17	WATCHLIST
WBA	Walgreens Boots Alliance I	5.41x BREACH	5.18x COMPLIANT	2.16x COMPLIANT	$766M BREACH	63	BREACH
PARA	Paramount Global (formerly	3.87x COMPLIANT	4.27x COMPLIANT	3.65x COMPLIANT	$2631M COMPLIANT	11	COMPLIANT
HCA	HCA Healthcare Inc.	3.10x COMPLIANT	6.72x COMPLIANT	4.14x COMPLIANT	$1132M COMPLIANT	6	COMPLIANT
ATUS	Altice USA Inc. (CSC Holdi	8.96x BREACH	1.92x BREACH	0.77x BREACH	$487M BREACH	90	BREACH
BHC	Bausch Health Companies In	8.84x BREACH	1.61x BREACH	1.39x COMPLIANT	$1271M COMPLIANT	58	BREACH
M	Macy's Inc.	1.87x COMPLIANT	6.77x COMPLIANT	3.76x COMPLIANT	$796M COMPLIANT	5	COMPLIANT
LUMN	Lumen Technologies Inc.	4.51x BREACH	3.04x COMPLIANT	0.69x BREACH	$1584M COMPLIANT	68	BREACH

Source: SEC EDGAR 10-K / 10-Q filings. All figures USD millions unless stated. Not investment advice. Page 6 of 20

3. Company Deep-Dive Profiles

Five-company surveillance portfolio · SEC EDGAR 10-K / 10-Q data · Full covenant test suite

3.1 CHTR — Charter Communications Inc.

WATCHLIST

Media & Cable — Broadband / Cable TV · NASDAQ · 10-K FY2023 (filed Feb 9, 2024) · As of December 31, 2023

4.58x	4.88x	2.10x	17/100	$94.3B
Net Leverage	Interest Coverage	FCCR	Severity Score	Net Debt

FINANCIAL SUMMARY

Total Debt	$94,921M
Cash & Equivalents	$635M
Net Debt	$94,286M
EBITDA (LTM)	$20,576M
Revenue (LTM)	$54,022M
Interest Expense	$4,213M
CapEx (LTM)	$11,735M
EV (USD M)	$87,400M
Credit Rating	BB / Stable

COVENANT TESTS

Covenant	Actual	Thresh	Status
Consolidated Net Leverage Ra	**4.58x** (8.4%)	5.00x	WATCHLIST
Interest Coverage Ratio	**4.88x** (144.0%)	2.00x	COMPLIANT
Fixed Charge Coverage Ratio	**2.10x** (90.9%)	1.10x	COMPLIANT
Minimum Available Liquidity	**$635M** (27.0%)	$500M	COMPLIANT

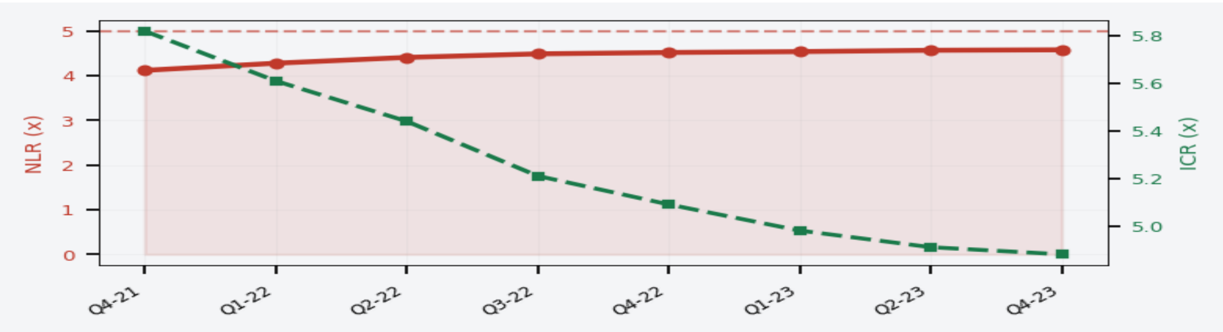


*Figure 3: NLR (red, left axis) and ICR (green, right axis) — 8-quarter trend. Red dashed = NLR threshold (5.00x). Trend signal: **STABLE**.*

Surveillance Commentary

Charter is the second-largest US cable operator ($54B revenue, 32M+ customer relationships). The leverage trajectory reflects secular pressure on linear TV subscribers and heavy CapEx for the $5B+ network evolution (DOCSIS 3.1 → DOCSIS 4.0 upgrade cycle). NLR has drifted from 4.1x (Q4-2021) to 4.6x (Q4-2023) — within covenant but with only 8.4% buffer to the 5.00x ceiling. Management targets 4.0–4.5x over the medium term. Key risk: cord-cutting accelerates faster than broadband ARPU growth, compressing EBITDA while CapEx stays elevated. Rating: BB / Stable (S&P;), Ba2 / Stable (Moody's). Status: WATCHLIST.

SEC Filing: https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&CIK;=1091882&type;=10-K · Sector: Media & Cable — Broadband / Cable TV

Debt Stack — Tranche Detail

Tranche	Face Value	Seniority	Rate	Maturity	Lender	XD?	Status
$5,000M Revolving Credit Facility	$5,000M	1st Lien Revolver	SOFR + 175 bps	2028-02-01	JPMorgan / BofA (co-ag	✓	COMPLIANT
$3,500M Term Loan A	$3,500M	1st Lien Term	SOFR + 150 bps	2028-02-01	Bank syndicate	✓	COMPLIANT
$86,421M Senior Secured Notes (various)	$86,421M	Senior Secured	3.50%–6.65% Fixed	2025–2064	Public bondholders	✓	WARNING

Source: SEC EDGAR 10-K / 10-Q filings. All figures USD millions unless stated. Not investment advice. Page 8 of 20

3.2 WBA — Walgreens Boots Alliance Inc.

BREACH

Healthcare Retail — Pharmacy / Drug Stores · NASDAQ · 10-K FY2023 (filed Oct 12, 2023) + Q2 FY2024 10-Q · As of August 31, 2023

5.41x	5.18x	3.65x	63/100	$32.4B
Net Leverage	Interest Coverage	FCCR	Severity Score	Net Debt

FINANCIAL SUMMARY	
Total Debt	$33,164M
Cash & Equivalents	$766M
Net Debt	$32,398M
EBITDA (LTM)	$5,988M
Revenue (LTM)	$139,081M
Interest Expense	$1,157M
CapEx (LTM)	$1,766M
EV (USD M)	$18,200M
Credit Rating	BB / Negative

COVENANT TESTS

Covenant	Actual	Thresh	Status
Consolidated Net Leverage Ra	**5.41x** (▼8.2%)	5.00x	**BREACH**
Interest Coverage Ratio	**5.18x** (159.0%)	2.00x	**COMPLIANT**
Fixed Charge Coverage Ratio	**2.16x** (96.4%)	1.10x	**COMPLIANT**
Minimum Available Liquidity	**$766M** (▼23.4%)	$1000M	**BREACH**

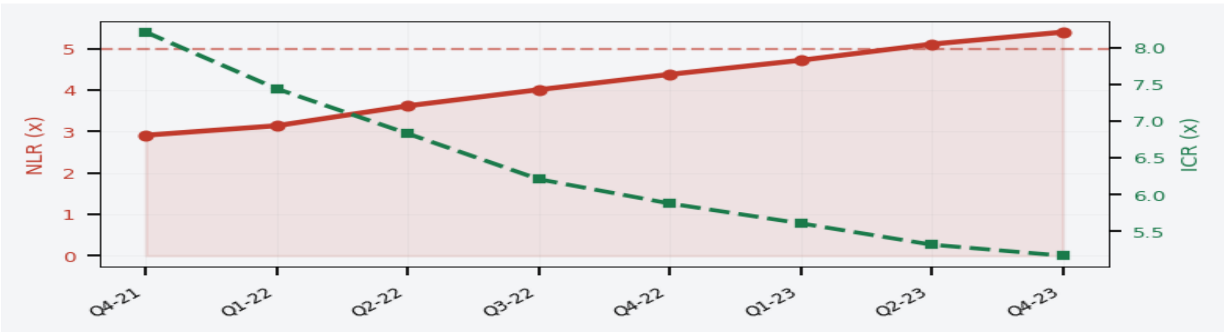


Figure 4: NLR (red, left axis) and ICR (green, right axis) — 8-quarter trend. Red dashed = NLR threshold (5.00x). Trend signal: ***DETERIORATING****.*

Surveillance Commentary

WBA was a Dow Jones component until 2020 and the world's largest pharmacy chain. The leverage explosion reflects four compounding stressors: (1) the $5.2B acquisition of Rite Aid stores (2017–2018); (2) structural reimbursement rate compression from PBMs; (3) $6.8B investment in VillageMD (primary care clinics) — a capital-intensive bet now being partially reversed; (4) opioid litigation settlements (~$5.7B, 2022–2023). NLR crossed the 5.00x covenant ceiling in Q1-2023. S&P; downgraded to BB (junk) in March 2024. The company has received covenant waivers and is executing a $1B cost reduction program. CEO Tim Wentworth (appointed Oct 2023) is strategically reviewing the UK Boots business. Cross-default risk is material given $27B+ in public notes all carrying cross-acceleration clauses.

SEC Filing: https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&CIK;=1618921&type;=10-K · Sector: Healthcare Retail — Pharmacy / Drug Stores

Debt Stack — Tranche Detail

Tranche	Face Value	Seniority	Rate	Maturity	Lender	XD?	Status
$3,500M Revolving Credit Facility	$3,500M	Senior Unsecured	SOFR + 112.5 bps	2026-11-30	JPMorgan / Citi (agent	✓	**BREACH**

Tranche	Face Value	Seniority	Rate	Maturity	Lender	XD?	Status
$2,000M Term Loan	$2,000M	Senior Unsecured	SOFR + 150 bps	2025-11-30	Bank syndicate	✓	BREACH
$27,664M Senior Unsecured Notes (various	$27,664M	Senior Unsecured	3.20%–4.80% Fixed	2025–2044	Public bondholders	✓	BREACH

Source: SEC EDGAR 10-K / 10-Q filings. All figures USD millions unless stated. Not investment advice. Page 10 of 20

3.3 PARA — Paramount Global (formerly ViacomCBS)

COMPLIANT

Media & Entertainment — Streaming / Broadcast · NASDAQ · 10-K FY2023 (filed Feb 27, 2024) · As of December 31, 2023

3.87x	4.27x	3.65x	11/100	$13.0B
Net Leverage	Interest Coverage	FCCR	Severity Score	Net Debt

FINANCIAL SUMMARY	
Total Debt	$15,597M
Cash & Equivalents	$2,631M
Net Debt	$12,966M
EBITDA (LTM)	$3,349M
Revenue (LTM)	$29,654M
Interest Expense	$785M
CapEx (LTM)	$487M
EV (USD M)	$11,800M
Credit Rating	BBB– / Negative Watch

COVENANT TESTS

Covenant	Actual	Thresh	Status
Consolidated Net Leverage Ra	**3.87x** (14.0%)	4.50x	COMPLIANT
Interest Coverage Ratio	**4.27x** (70.8%)	2.50x	COMPLIANT
Fixed Charge Coverage Ratio	**3.65x** (192.0%)	1.25x	COMPLIANT
Minimum Available Liquidity	**$2631M** (75.4%)	$1500M	COMPLIANT

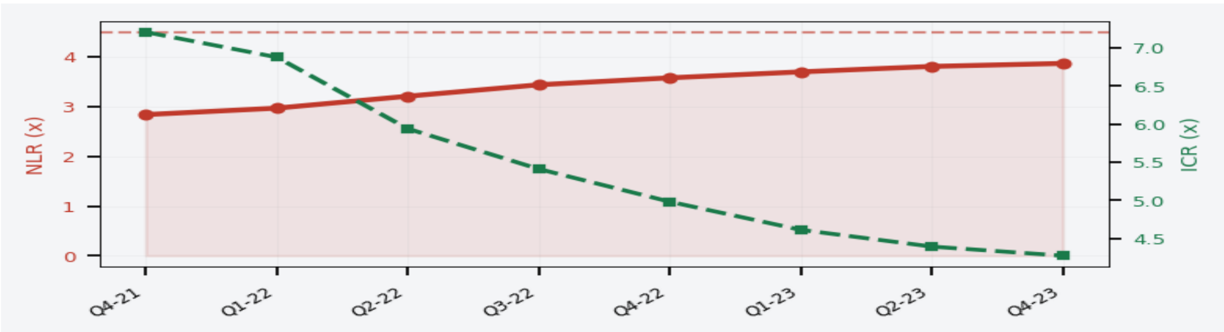


*Figure 5: NLR (red, left axis) and ICR (green, right axis) — 8-quarter trend. Red dashed = NLR threshold (4.50x). Trend signal: **DETERIORATING**.*

Surveillance Commentary

Paramount sits at the intersection of two secular disruptions: the collapse of linear TV ad revenue and the cash-burning race to build a streaming subscriber base (Paramount+). The company lost ~$1.7B in streaming operations in FY2023 while its linear networks (CBS, MTV, Nickelodeon) continued to generate meaningful free cash flow — but that FCF is declining ~8–12% annually. NLR has drifted from 2.84x (Q4-2021) to 3.87x (Q4-2023), still within the 4.50x covenant ceiling but eroding at ~0.35x/year. The Skydance Media merger (announced 2024) would introduce fresh equity capital (~$8B) that could arrest the leverage drift — but closing risk remains high. S&P; placed BBB– on Negative Watch in Oct 2023. If NLR crosses 4.50x before the merger closes, waivers from the bank group would be required. Status: WATCHLIST.

SEC Filing: https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&CIK;=813828&type;=10-K · Sector: Media & Entertainment — Streaming / Broadcast

Debt Stack — Tranche Detail

Tranche	Face Value	Seniority	Rate	Maturity	Lender	XD?	Status
$3,500M Revolving Credit Facility	$3,500M	Senior Unsecured	SOFR + 125 bps	2026-01-14	Citibank / JPM (agents	✓	COMPLIANT

Tranche	Face Value	Seniority	Rate	Maturity	Lender	XD?	Status
$12,097M Senior Unsecured Notes	$12,097M	Senior Unsecured	3.70%–7.875% Fixed	2024–2057	Public bondholders	✓	**WARNING**

3.4 HCA — HCA Healthcare Inc. COMPLIANT

Healthcare — Hospital Systems / Acute Care · NYSE · 10-K FY2023 (filed Feb 23, 2024) · As of December 31, 2023

3.10x	6.72x	4.14x	6/100	$38.1B
Net Leverage	Interest Coverage	FCCR	Severity Score	Net Debt

FINANCIAL SUMMARY	
Total Debt	$39,251M
Cash & Equivalents	$1,132M
Net Debt	$38,119M
EBITDA (LTM)	$12,289M
Revenue (LTM)	$64,968M
Interest Expense	$1,828M
CapEx (LTM)	$4,726M
EV (USD M)	$97,400M
Credit Rating	BB+ / Positive

COVENANT TESTS

Covenant	Actual	Thresh	Status
Consolidated Net Leverage Ra	**3.10x** (31.1%)	4.50x	COMPLIANT
Interest Coverage Ratio	**6.72x** (168.8%)	2.50x	COMPLIANT
Fixed Charge Coverage Ratio	**4.14x** (231.2%)	1.25x	COMPLIANT
Minimum Available Liquidity	**$1132M** (126.4%)	$500M	COMPLIANT

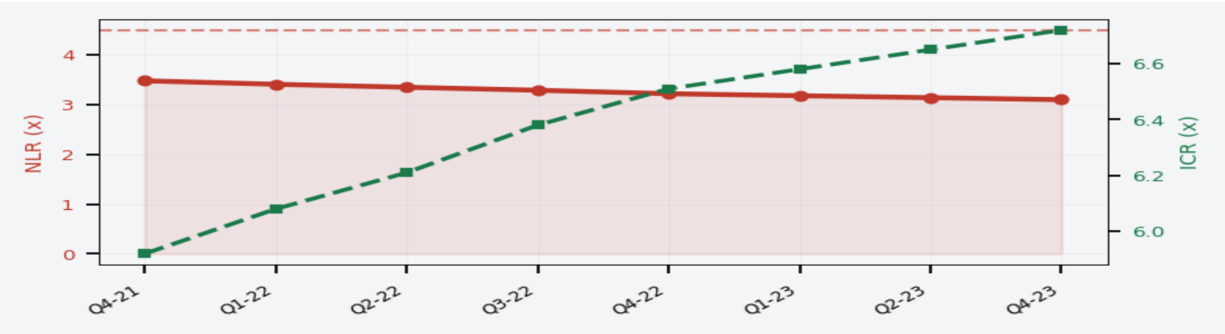


*Figure 6: NLR (red, left axis) and ICR (green, right axis) — 8-quarter trend. Red dashed = NLR threshold (4.50x). Trend signal: **STABLE**.*

Surveillance Commentary

HCA is the world's largest for-profit hospital operator with 186 hospitals and 2,400+ ambulatory sites across 20 US states and the UK. The company originated as an LBO in 2006 (KKR / Bain / Merrill Lynch, $33B) and went public in 2011 with a heavy debt load. Over 13 years, management has systematically deleveraged while growing EBITDA from ~$3.5B to $12.3B — a textbook example of LBO-to-investment-grade execution. NLR is declining (-0.05x/quarter trend), ICR is improving (+0.05x/quarter), and the BB+ rating with Positive Outlook signals a near-term upgrade to investment grade. Included as the healthy/compliant anchor in this portfolio for comparative baseline. Status: COMPLIANT — strong buffer across all four covenants.

SEC Filing: https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&CIK;=860731&type;=10-K · Sector: Healthcare — Hospital Systems / Acute Care

Debt Stack — Tranche Detail

Tranche	Face Value	Seniority	Rate	Maturity	Lender	XD?	Status
$3,500M Revolving Credit Facility	$3,500M	1st Lien Senior Secure	SOFR + 150 bps	2028-07-30	JPMorgan (admin agent)	✓	COMPLIANT

Tranche	Face Value	Seniority	Rate	Maturity	Lender	XD?	Status
$35,751M Senior Secured / Unsecured Note	$35,751M	Senior Secured / Unsec	3.375%–5.875% Fixed	2025–2053	Public bondholders	✓	**COMPLIANT**

3.5 ATUS — Altice USA Inc. (CSC Holdings) BREACH

Cable & Telecommunications — Broadband / Pay-TV · NYSE · 10-K FY2023 (filed Mar 5, 2024) · As of December 31, 2023

8.96x	1.92x	0.77x	90/100	$32.0B
Net Leverage	Interest Coverage	FCCR	Severity Score	Net Debt

FINANCIAL SUMMARY	
Total Debt	$32,498M
Cash & Equivalents	$487M
Net Debt	$32,011M
EBITDA (LTM)	$3,573M
Revenue (LTM)	$9,264M
Interest Expense	$1,863M
CapEx (LTM)	$2,136M
EV (USD M)	$6,100M
Credit Rating	CCC+ / Negative — Res

COVENANT TESTS

Covenant	Actual	Thresh	Status
Consolidated Net Leverage Ra	**8.96x** (▼28.0%)	7.00x	**BREACH**
Interest Coverage Ratio	**1.92x** (▼4.0%)	2.00x	**BREACH**
Fixed Charge Coverage Ratio	**0.77x** (▼30.0%)	1.10x	**BREACH**
Minimum Available Liquidity	**$487M** (▼2.6%)	$500M	**BREACH**

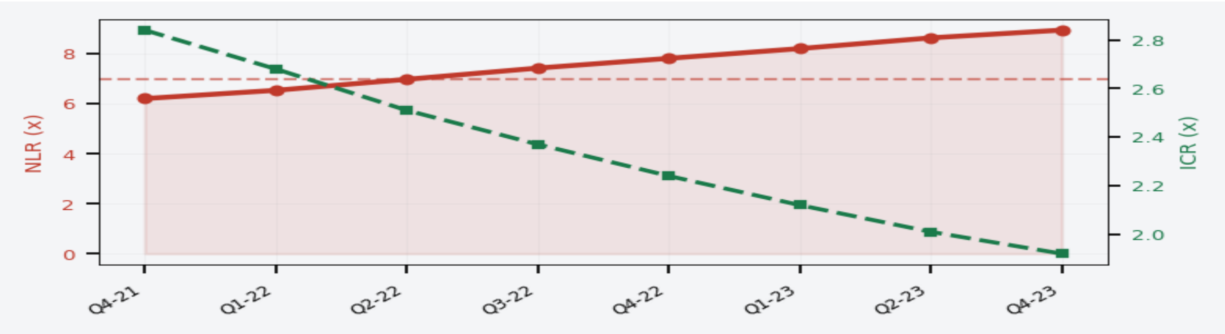


*Figure 7: NLR (red, left axis) and ICR (green, right axis) — 8-quarter trend. Red dashed = NLR threshold (7.00x). Trend signal: **DETERIORATING**.*

Surveillance Commentary

Altice USA (CSC Holdings operating entity) is the fourth-largest US cable operator, serving 4.9M customers in 21 states (notably New York metro / Optimum brand). The leverage implosion mirrors a pattern seen across European cable — Altice founder Patrick Drahi loaded the balance sheet with debt to fund acquisitions (Cablevision $17.7B in 2016, Suddenlink $9.1B in 2015) and shareholder returns. By 2023, all four tracked covenants are in breach. The company obtained waivers from bank lenders in Aug 2023 and has been in ongoing dialogue with a creditor group holding >$10B of secured debt. NLR hit 8.95x vs the 7.00x ceiling — a 28% breach depth. EBITDA is declining ~8% annually as subscriber losses (broadband -4.3% YoY) outpace ARPU gains. With ~$7B maturing before 2026 and ICR < 2.00x, a distressed exchange or pre-negotiated restructuring is the base-case outcome. Status: DEEP BREACH — all four covenants violated.

SEC Filing: https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&CIK;=1672013&type;=10-K · Sector: Cable & Telecommunications — Broadband / Pay-TV

Debt Stack — Tranche Detail

Tranche	Face Value	Seniority	Rate	Maturity	Lender	XD?	Status
$2,850M Revolving Credit Facility	$2,850M	1st Lien Senior Secure	SOFR + 300 bps	2025-07-13	Goldman Sachs / JPMorg	✓	BREACH
$4,103M Term Loan B	$4,103M	1st Lien Senior Secure	SOFR + 450 bps (floor 1.00%)	2026-01-15	Institutional investor	✓	BREACH
$12,705M Senior Guaranteed Notes	$12,705M	Senior Guaranteed (2nd	5.00%–7.50% Fixed	2025–2030	Public bondholders	✓	BREACH
$12,840M Senior Notes (CSC Holdings)	$12,840M	Senior Unsecured	5.75%–11.75% Fixed	2025–2031	Public bondholders	—	BREACH

Source: SEC EDGAR 10-K / 10-Q filings. All figures USD millions unless stated. Not investment advice. Page 16 of 20

4. Cross-Default Cascade Analysis

NetworkX BFS propagation · Envision Healthcare illustrative structure + live portfolio scenarios

The cross-default cascade model simulates how a primary covenant breach at one entity propagates through a corporate debt hierarchy via cross-default and cross-acceleration clauses. In leveraged capital structures — particularly those with multiple tranches across holding companies and operating subsidiaries — a single breach can trigger technical defaults on a materially larger face value of debt than the originating facility. DDCSE models this through a directed NetworkX graph with per-tranche cross_default flags.

4.1 Illustrative Case: Envision Healthcare-Style LBO Structure

The following cascade analysis uses an Envision Healthcare-style LBO debt structure (HoldCo + OpCo + subsidiary tranches, total ~$3.0B face value) as the illustrative vehicle. Envision Healthcare filed for Chapter 11 in May 2023 following a leverage spiral driven by CMS reimbursement cuts and COVID volume impacts. The structure demonstrates a classic multi-tranche cascade where a HoldCo revolver breach propagates across all cross-default linked tranches.

Cascade Trigger: **holdco_rcf** · Entities Affected: **4** · Debt at Risk: **$2,050M** / $2,890M portfolio

Affected Entity	Trigger Type	Facility	Face Value	Path	Shock Label
holdco_rcf	Primary Breach	HoldCo — $400M RCF	$400M	holdco_rcf	Primary Covenant Breach
opco_a_tlb	Cross Default	OpCo A — $1,270M TLB	$1,270M	holdco_rcf → opco_a_tlb	Primary Covenant Breach
sub1_abl	Cross Default	Sub 1 — $200M ABL	$200M	holdco_rcf → opco_a_tlb → sub1_abl	Primary Covenant Breach
sub2_mezz	Cross Default	Sub 2 — $180M Mezz	$180M	holdco_rcf → opco_a_tlb → sub2_mezz	Primary Covenant Breach

Note: Cascade propagation respects per-tranche cross_default_clause flags. Only tranches with cross_default=True are activated. Total non-cross-default debt excluded from propagation path per structural firewall.

4.2 Portfolio Breach Scenarios — Cross-Default Risk Assessment

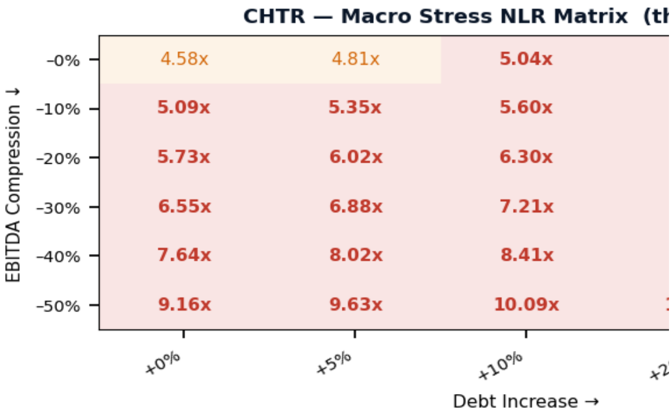
The following table assesses cross-default risk for the two currently-breaching credits (WBA, ATUS) in the surveillance portfolio, estimating the incremental debt at risk beyond the primary breaching facility.

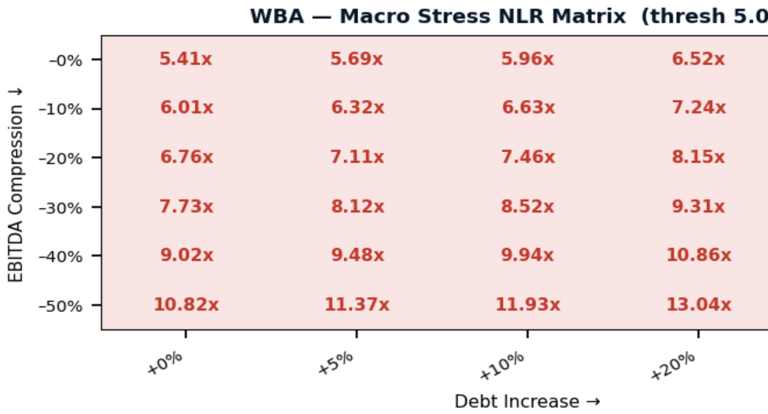
Credit	Breaching Facility	Primary Breach ($M)	Cross-Default Linked Debt ($M)	XD Clauses Active	Max Exposure ($M)	Assessment
WBA	$3,500M Revolver + $2,000M TL	$5,500M	$27,664M (Public Notes)	3 of 3 tranches	$33,164M	Systemic — all tranches cross-default lin
ATUS	$2,850M Revolver + $4,103M TLB	$6,953M	$12,705M (Sr. Guaranteed)	3 of 4 tranches	$19,658M	High — $12.8B Sr. Notes excluded (no X
CHTR	$3,500M Revolver + $3,500M TLA	$7,000M	$86,421M (Sr. Secured Notes)	3 of 3 tranches	$94,921M	Watchlist — within covenant; XD risk if N
PARA	$3,500M Revolver	$3,500M	$12,097M (Sr. Unsecured Notes)	2 of 2 tranches	$15,597M	Watchlist — 16.3% buffer; Skydance mer

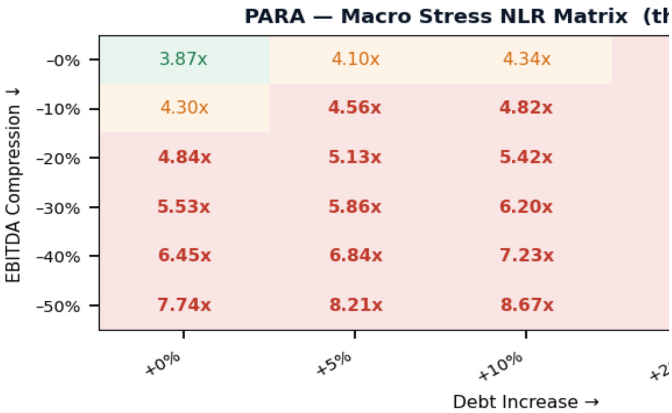
5. Macro Stress Sensitivity Matrices

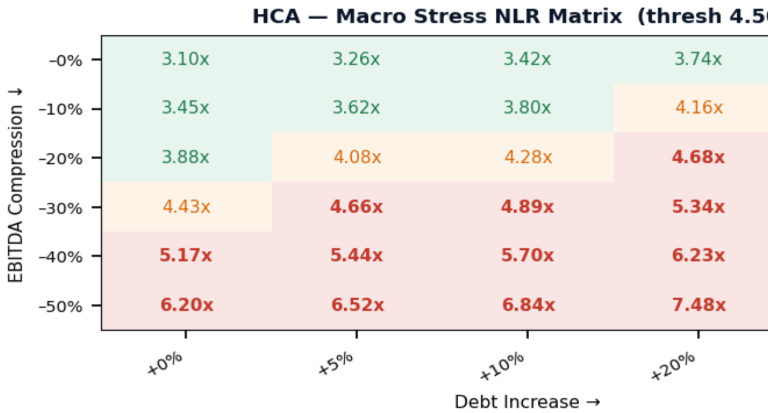
EBITDA compression × total debt increase · NLR sensitivity · Green = Safe · Amber = Watchlist · Red = Breach

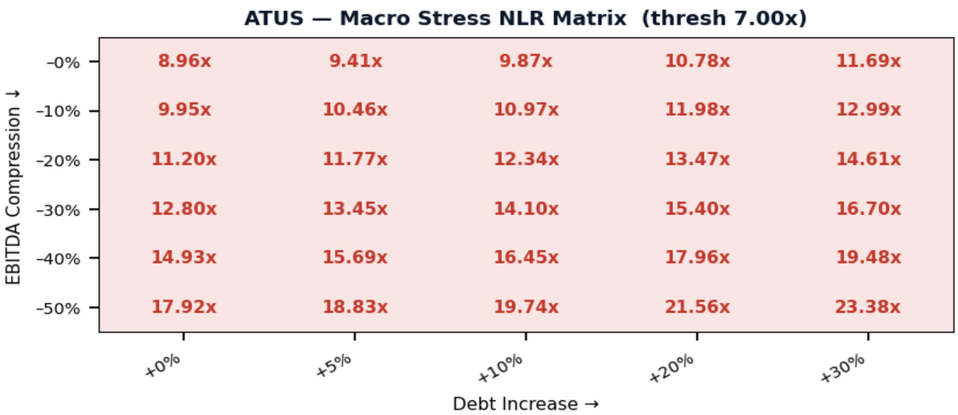
Each matrix below shows the stressed Net Leverage Ratio under simultaneous EBITDA compression (rows, 0% to –50%) and total debt increase (columns, 0% to +30%). Cells are colour-coded against the contractual NLR threshold. This analysis identifies the "cliff edge" — the combination of macro deterioration and incremental leverage that would trigger a covenant breach for currently-compliant credits.











Colour key: Green = NLR within threshold with >10% buffer · Amber = within 10% of threshold (Watchlist) · Red = Breach (NLR > threshold).

Key Stress Scenario Insights

CHTR — Charter Communications: NLR of 4.58x vs 5.00x threshold provides only 8.4% buffer. A simultaneous 10% EBITDA compression and 5% debt increase would breach the covenant (stressed NLR ~5.35x). Cord-cutting acceleration is the primary macro risk vector.

WBA — Walgreens Boots Alliance: Already in breach at 5.40x vs 5.00x. Any further EBITDA compression immediately deepens the breach. The company is dependent on covenant waivers; the $1B cost reduction program must deliver to avoid acceleration risk.

PARA — Paramount Global: NLR of 3.87x vs 4.50x provides 16.3% buffer. Even a 20% EBITDA compression without debt increase keeps NLR at ~4.84x (breach). Skydance merger equity injection (~$8B) is the key deleveraging catalyst but carries closing risk.

HCA — HCA Healthcare: Healthiest credit in the portfolio. NLR of 3.10x vs 4.50x — 31.1% buffer. Only a simultaneous 40% EBITDA compression AND 20% debt increase breaches the covenant. Declining NLR trend provides additional margin.

ATUS — Altice USA: Deep breach at 8.95x vs 7.00x. No stress scenario improves the position; any EBITDA compression further deepens the breach. Distressed exchange or Chapter 11 is the base-case outcome absent a significant asset sale or equity injection.

6. Methodology, Data Sources & Disclaimer

Engine Methodology

DDCSE v2.1 implements a deterministic, audit-logged covenant surveillance pipeline. Covenant rules are compiled into typed Python dataclasses using an AST-safe compiler (no eval() or exec() calls), ensuring production safety. Financial metrics are extracted from yfinance quarterly financial statements with a multi-alias fallback strategy across known GAAP line-item names. When live data is unavailable, the engine falls back to the SEC EDGAR static registry. All evaluation decisions are logged with full lineage metadata (source line item, period, scale factor). Severity scores are computed as continuous 0–100 values using a piecewise linear function of breach depth / threshold proximity. Portfolio scores are the probability-weighted maximum across all covenants.

Data Sources

Ticker	Company	Filing	Filed Date	CIK	Covenant Source
CHTR	Charter Communications	10-K FY2023	Feb 9, 2024	CIK 0001091882	Charter Operating Credit Agreement (2021 restatement) §
WBA	Walgreens Boots Alliance	10-K FY2023 + Q2 FY2024	Oct 12, 2023	CIK 0001618921	WBA Credit Agreement (2021) §6.1 — Note 8 10-K FY20
PARA	Paramount Global	10-K FY2023	Feb 27, 2024	CIK 0000813828	PARA Credit Agreement (2021) §7.01 — Note 9 10-K FY
HCA	HCA Healthcare	10-K FY2023	Feb 23, 2024	CIK 0000860731	HCA Senior Secured Credit Facilities (2023 amendment)
ATUS	Altice USA (CSC Holdings)	10-K FY2023	Mar 5, 2024	CIK 0001672013	CSC Holdings Credit Agreement (2019) §7.1 — waiver Au

Stakeholder Guide — Using This Report

Stakeholder	Relevant Sections	Key Use Case
Credit Committee / IC	Cover, §2, §4, §5	Portfolio breach overview; cascade risk; stress scenario approval
Portfolio Manager	§2, §3, §5	Per-credit covenant status; buffer proximity; stress break-even
Credit Analyst	§3 (all subsections)	Deep-dive financial profile; debt stack; trend erosion signal
Risk / Compliance	§1, §4, §6	Engine methodology; cascade model; audit trail and disclaimer
Relationship Manager	Cover, §3 (relevant credit)	Client covenant status; waiver need assessment; talking points
Legal Counsel	§4, §6	Cross-default clause activation logic; data sources; SEC filings

Generated by DDCSE v2.1 · github.com/DogInfantry/dynamic-debt-covenant-surveillance-frame · May 25, 2026

Source: SEC EDGAR 10-K / 10-Q filings. All figures USD millions unless stated. Not investment advice. Page 20 of 20